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Lillo et al.

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(54) **STRAWBERRY PLANT NAMED ‘DRISSTRAWONEHUNDREDFOUR’**

(50) Latin Name: ***Fragaria* x *ananassa***
Varietal Denomination: **DrisStraw OneHundredFour**

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None
See application file for complete search history.

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(57) **ABSTRACT**

A new and distinct variety of strawberry plant named ‘DrisStrawOneHundredFour’, particularly selected for its yield potential and shelf life, as well as its upright growth habit, long and well exposed trusses, and fruit quality, is disclosed.

5 Drawing Sheets

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STRAWBERRY PLANT NAMED 'DRISSTRAWONEHUNDREDFOUR'

Latin name: Botanical classification: *Fragaria* x *ananassa.*

Varietal denomination: The varietal denomination of the claimed variety of strawberry plant is 'DrisStrawOneHundredFour'.

BACKGROUND OF THE INVENTION

Cultivated strawberry is a hybrid species of the genus *Fragaria* that is grown worldwide for its fruit. Modern strawberry was first bred in Brittany, France, in the 18th century by crossing *Fragaria virginiana* with *Fragaria chiloensis*. Strawberry fruit is an aggregate accessory fruit, with the fleshy part of the fruit being derived from the receptacle that holds the ovaries.

Strawberry varieties vary widely in color, size, shape, flavor, season of ripening, degree of fertility, and susceptibility to disease. Certain varieties vary in foliage, and some vary in the relative development of their reproductive organs. Typically, strawberry flowers appear hermaphroditic in structure, but function as either male or female. Generally, commercial production of strawberry plants involves propagation from runners and distribution as either plugs or bare root plants. Cultivation is either perennial or annual plasticulture. During the off season, strawberries can also be produced in greenhouses.

Strawberry fruit is widely appreciated for its characteristic bright red color, aroma, juicy texture, and sweetness. Strawberry fruit is a popular fruit that is generally consumed either fresh or in prepared foods, such as preserves and baked goods.

Strawberry is an important and valuable fruit crop. Accordingly, there is a need for new varieties of strawberry plants. In particular, there is a need for improved varieties of strawberry plant that are stable, high yielding, and agronomically sound.

SUMMARY OF THE INVENTION

In order to meet these needs, the present invention is directed to an improved variety of strawberry plant. In particular, the invention relates to a new and distinct variety of strawberry plant (*Fragaria* x *ananassa*), which has been denominated as 'DrisStrawOneHundredFour'.

Strawberry plant variety 'DrisStrawOneHundredFour' was selected in East Malling, Kent, United Kingdom in July 2017 and originated from a controlled cross between the proprietary strawberry female parent 'ADUKE 146-001' (unpatented) and the proprietary strawberry male parent 'WUKE 085-001' (unpatented). The original seedling of 'DrisStrawOneHundredFour' was first asexually propagated via stolons in East Malling, Kent, United Kingdom in July 2017.

'DrisStrawOneHundredFour' was subsequently asexually propagated via stolons, and has undergone testing at test plots in East Malling, Kent, United Kingdom for four years (2017 to 2021). The present variety has been found to be stable and reproduce true to type through successive asexual propagations via stolons and tissue culture.

'DrisStrawOneHundredFour' was particularly selected for its yield potential and shelf life, as well as its upright growth habit, long and well exposed trusses, and fruit quality.

DESCRIPTION OF THE DRAWINGS

This new strawberry plant is illustrated by the accompanying photographs. The colors shown are as true as can be reasonably obtained by conventional photographic procedures. Unless otherwise indicated, the photographs are of plants that are nine to eleven months old.

FIG. **1** illustrates whole fruit of variety 'DrisStrawOneHundredFour'.

FIG. **2** illustrates longitudinal sections of fruit of variety 'DrisStrawOneHundredFour'.

FIG. **3** illustrates the lower surface (top row) and upper surface (bottom row) of flowers of variety 'DrisStrawOneHundredFour'.

FIG. **4** illustrates the upper surface (top) and lower surface (bottom) of leaves of variety 'DrisStrawOneHundredFour'.

FIG. **5** illustrates a side view of whole plants of variety 'DrisStrawOneHundredFour'.

DETAILED BOTANICAL DESCRIPTION

The following detailed descriptions set forth the distinctive characteristics of 'DrisStrawOneHundredFour'. The data which define these characteristics is based on observations taken in East Malling, Kent, United Kingdom from 2017 to 2021. This description is in accordance with UPOV terminology. Color designations, color descriptions, and other phenotypical descriptions may deviate from the stated values and descriptions depending upon variation in environmental, seasonal, climatic, and cultural conditions. 'DrisStrawOneHundredFour' has not been observed under all possible environmental conditions. The botanical description of 'DrisStrawOneHundredFour' was taken from plants that were nine to eleven months old. The indicated values represent averages calculated from measurements of several plants. Color references are primarily to the RHS Colour Chart of The Royal Horticultural Society of London (RHS) (2015 edition). Descriptive terminology follows the *Plant Identification Terminology, An Illustrated Glossary,* 2nd edition by James G. Harris and Melinda Woolf Harris, unless where otherwise defined.

Classification:
- *Species.*—*Fragaria* x *ananassa.*
- *Common name.*—Strawberry.
- *Denomination.*—'DrisStrawOneHundredFour'.

Parentage:
- *Female parent.*—Proprietary strawberry plant 'ADUKE 146-001' (unpatented).
- *Male parent.*—Proprietary strawberry plant 'WUKE 085-001' (unpatented).

Plant:
- *Height.*—41.9 cm.
- *Width.*—36.8 cm.
- *Height/width ratio.*—1.14.
- *Number of crowns per plant.*—3.6.
- *Growth habit.*—Upright.
- *Density of foliage.*—Medium.
- *Vigor.*—Strong.

Stolon:
- *Average number of daughter plants per square foot.*—7.43.
- *Diameter (at bract).*—3.71 mm.
- *Overall color.*—RHS 144A (Strong yellow green).
- *Anthocyanin coloration.*—Very weak.
- *Density of pubescence.*—Sparse.

Fruiting truss:

Length (*from crown to base of terminal flower or fruit*).—53.1 cm.

Diameter (*at base of truss*).—5.5 mm.

Number of berries per truss.—7.3.

Attitude at first picking.—Prostrate.

Color (*at base of truss*).—RHS 144B (Strong yellow green).

Leaf:

Number of leaflets.—Three only.

Color of leaf upper surface.—RHS 136A (Dark green).

Color of leaf lower surface.—RHS 137C (Moderate yellow green).

Blistering.—Absent or weak.

Glossiness.—Medium.

Variegation.—Absent.

Terminal leaflet.—Length: 13.1 cm. Width: 8.9 cm. Length/width ratio: 1.5. Number of teeth per terminal leaflet: 25.1. Overall shape: Orbicular. Shape of base: Obtuse. Shape of apex: Rounded. Margin: Serrate to crenate. Margin profile: Flat (level with the leaflet blade). Shape in cross section: Straight.

Petiole.—Length: 27.6 cm. Diameter: 4.9 mm. Overall color: RHS 144B (Strong yellow green). Pubescence: Sparse. Attitude of hairs: Slightly outwards. Bract frequency (number present on each petiole): 1.4.

Petiolule.—Length: 15.2 mm. Diameter: 3 mm. Color: RHS 144B (Strong yellow green).

Stipule.—Length: 32.5 mm. Width: 8.7 mm. Stipule color: RHS 144A (Strong yellow green). Anthocyanin coloration: Medium. Anthocyanin color: RHS 51B (Deep pink). Pubescence: Absent or very sparse.

Inflorescence:

Number of flowers per plant.—24.

Position of inflorescence in relation to foliage.—Above.

Flowering interval.—Mid-April to mid-September.

Pedicel.—Attitude of hairs: Upwards.

Flower.—Flower diameter (petal tip to petal tip on non-flattened flower): 26.9 mm. Arrangement of petals: Overlapping. Size of calyx in relation to corolla: Larger. Receptacle color: RHS 144A (Strong yellow green). Stamen: Present. Anther color: RHS 13A (Vivid yellow).

Petal.—Length: 11.3 mm. Width: 11.3 mm. Length/width ratio: 1. Number of petals per flower: 5.9. Color of upper surface: RHS NN155B (White). Color of lower surface: RHS NN155B (White). Overall shape: Orbicular. Shape of apex: Rounded. Margin: Entire. Shape of base: Concavo-convex.

Calyx.—Diameter (sepal tip to sepal tip, measured on back of flower): 36.5 mm.

Sepal.—Length: 11.9 mm. Width: 5.2 mm. Number of sepals per flower: 12.3. Overall shape: Ovate. Shape of apex: Convex. Margin: Entire.

Fruit:

Fruit size.—Length: 52.8 mm. Width: 43 mm. Length/width ratio: 1.2.

Fruit hollow.—Length: 17.4 mm. Width: 4.8 mm. Length/width ratio: 3.6.

Shape.—Cordate.

Difference in shape of terminal and other fruits.—None or very slight.

Fruit color.—RHS N45A (Moderate red).

Evenness of color.—Even or very slightly uneven.

Glossiness.—Medium.

Evenness of surface.—Even or very slightly uneven.

Width of band without achenes.—Narrow.

Position of achenes.—Level with surface.

Position of calyx attachment.—Inserted.

Attitude of sepals.—Outwards.

Diameter of calyx in relation to diameter of fruit.—Slightly larger.

Adherence of calyx.—Strong.

Firmness.—Medium firm.

Color of flesh (*excluding core*).—RHS 34B (Vivid reddish orange).

Evenness of flesh color.—Even.

Distribution of flesh color.—Marginal and central.

Color of core.—RHS 35B (Moderate reddish orange).

Sweetness/soluble solids (*in ° Brix*).—7.8.

Individual fruit weight.—17.6 g/fruit.

Achenes.—Number of achenes per fruit: 345.8. Weight: 0.000587 g/achene. Color of upper (sunward) side: RHS 46A (Vivid red). Color of lower (shaded) side: RHS 2B (Brilliant greenish yellow).

Fruiting.—Harvest interval: Late May to late October. Type of bearing: Day neutral. Productivity: 1.250 kg to 1.544 kg of fruit per plant per season from nine to eleven-month-old plants when grown in East Malling, Kent, United Kingdom.

Resistance to abiotic stress, pests, and diseases:

Heat.—Moderately resistant.

Botrytis fruit rot (*Botrytis cinerea*).—Moderately susceptible.

Powdery mildew (*Podosphaera macularis*).—Resistant.

Verticillium wilt (*Verticillium dahliae*).—Moderately susceptible.

COMPARISON WITH PARENTAL AND REFERENCE VARIETIES

'DrisStrawOneHundredFour' differs from the female parent proprietary strawberry plant 'ADUKE 146-001' (unpatented) in that 'DrisStrawOneHundredFour' has significantly improved fruit shelf life compared to 'ADUKE 146-001'.

'DrisStrawOneHundredFour' differs from the male parent proprietary strawberry plant 'WUKE 085-001' (unpatented) in that 'DrisStrawOneHundredFour' has much higher yield potential as tray plants on table tops compared to 'WUKE 085-001'.

'DrisStrawOneHundredFour' differs from the reference variety 'DrisStrawFiftyEight' (U.S. Plant Pat. No. 30,851) in that 'DrisStrawOneHundredFour' has very weak stolon anthocyanin coloration, the number of stolons is medium, the shape of base on terminal leaflet is obtuse, and the width of band without achenes on fruit is narrow, whereas 'DrisStrawFiftyEight' has medium stolon anthocyanin coloration, the number of stolons is many, the shape of base on terminal leaflet is acute, and the width of band without achenes on fruit is absent or very narrow.

'DrisStrawOneHundredFour' differs from the reference variety 'DrisStrawTwo' (U.S. Plant Pat. No. 18,878) in that 'DrisStrawOneHundredFour' has medium density of foliage, very weak stolon anthocyanin coloration, the difference in shape of terminal and other fruits is none or very slight, and the position of calyx attachment is inserted, whereas 'DrisStrawTwo' has sparse density of foliage, strong stolon anthocyanin coloration, the difference in shape of terminal and other fruits is moderate, and the position of calyx attachment is raised.

We claim:

1. A new and distinct variety of strawberry plant named 'DrisStrawOneHundredFour' as shown and described herein.

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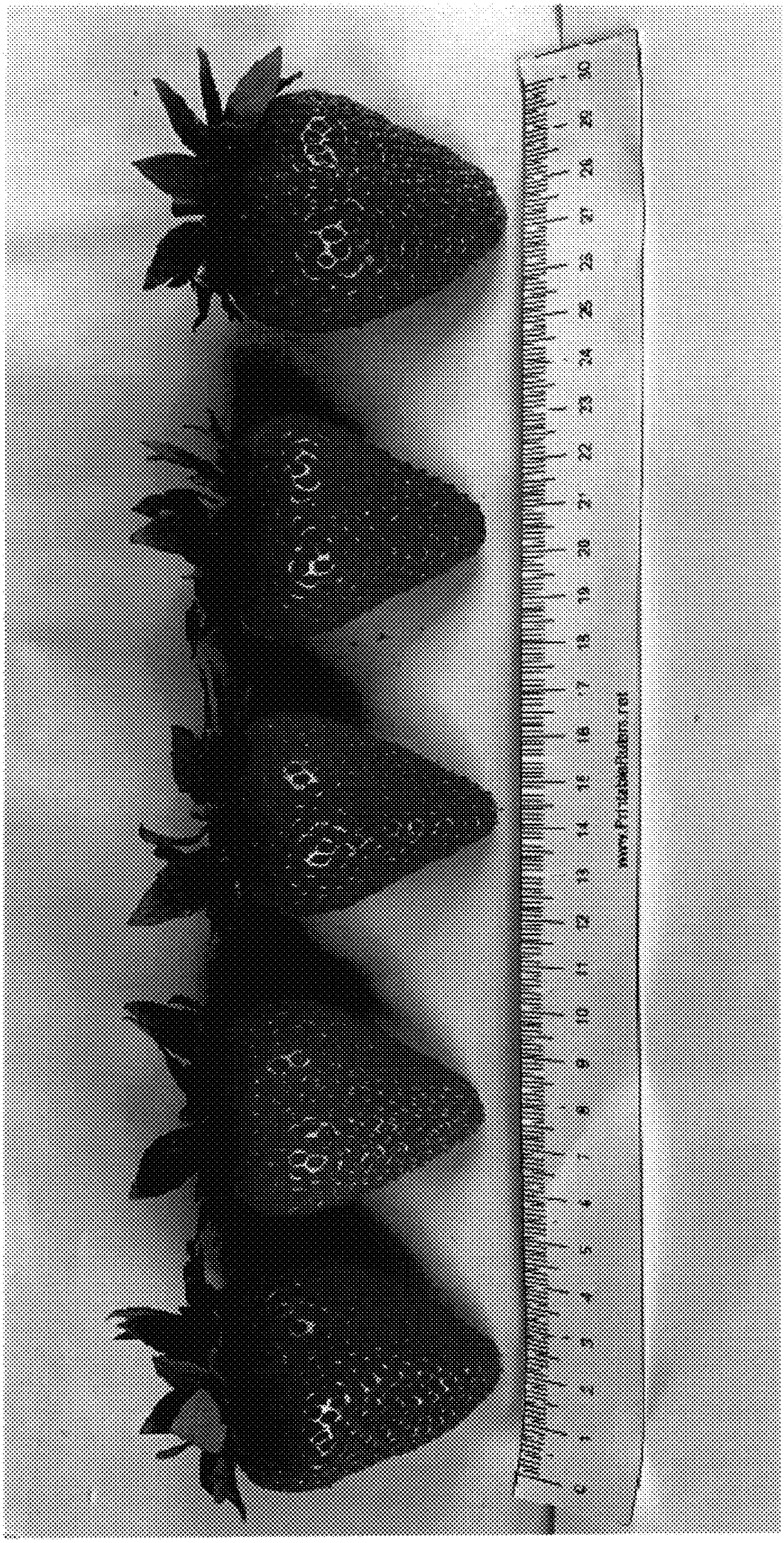

FIG. 1

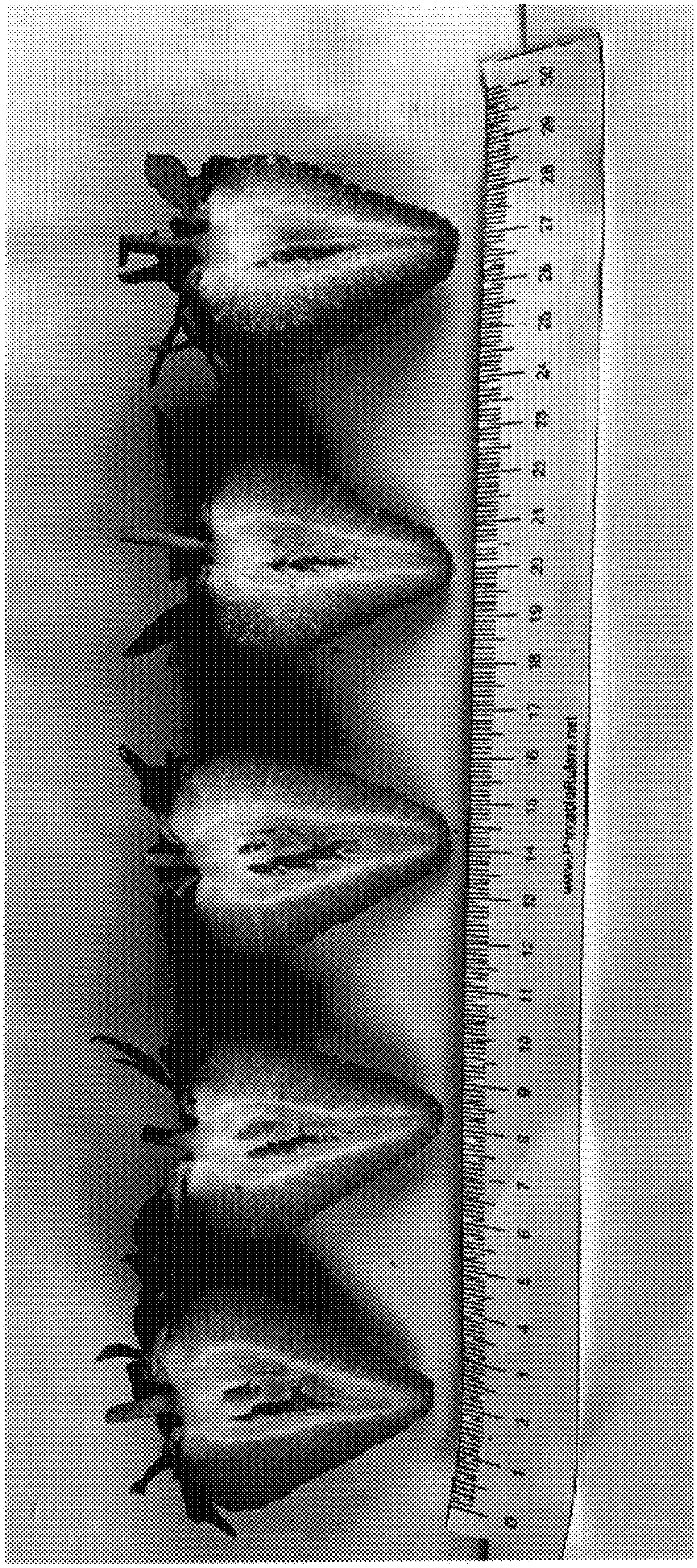

FIG. 2

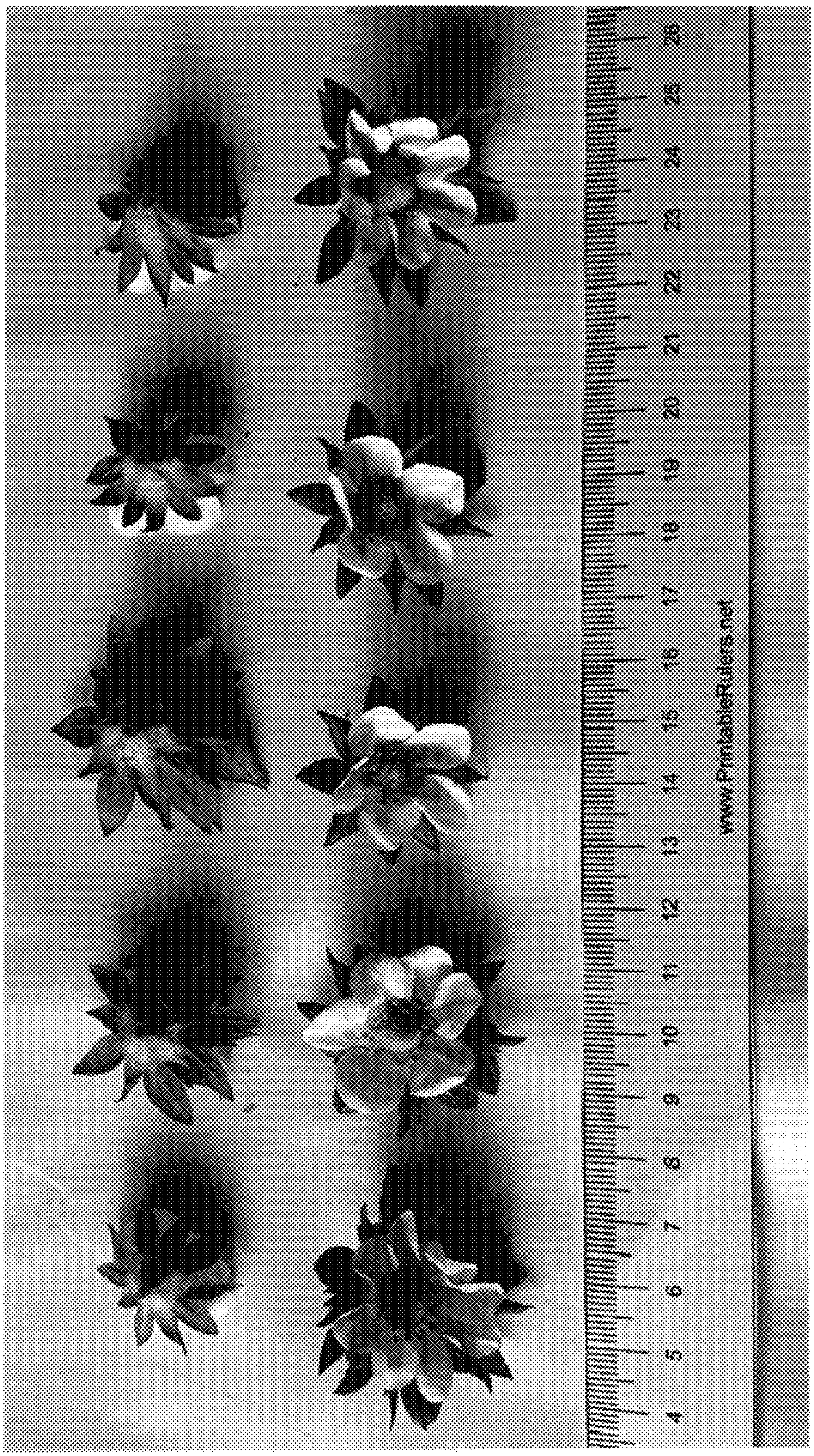
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FIG. 3

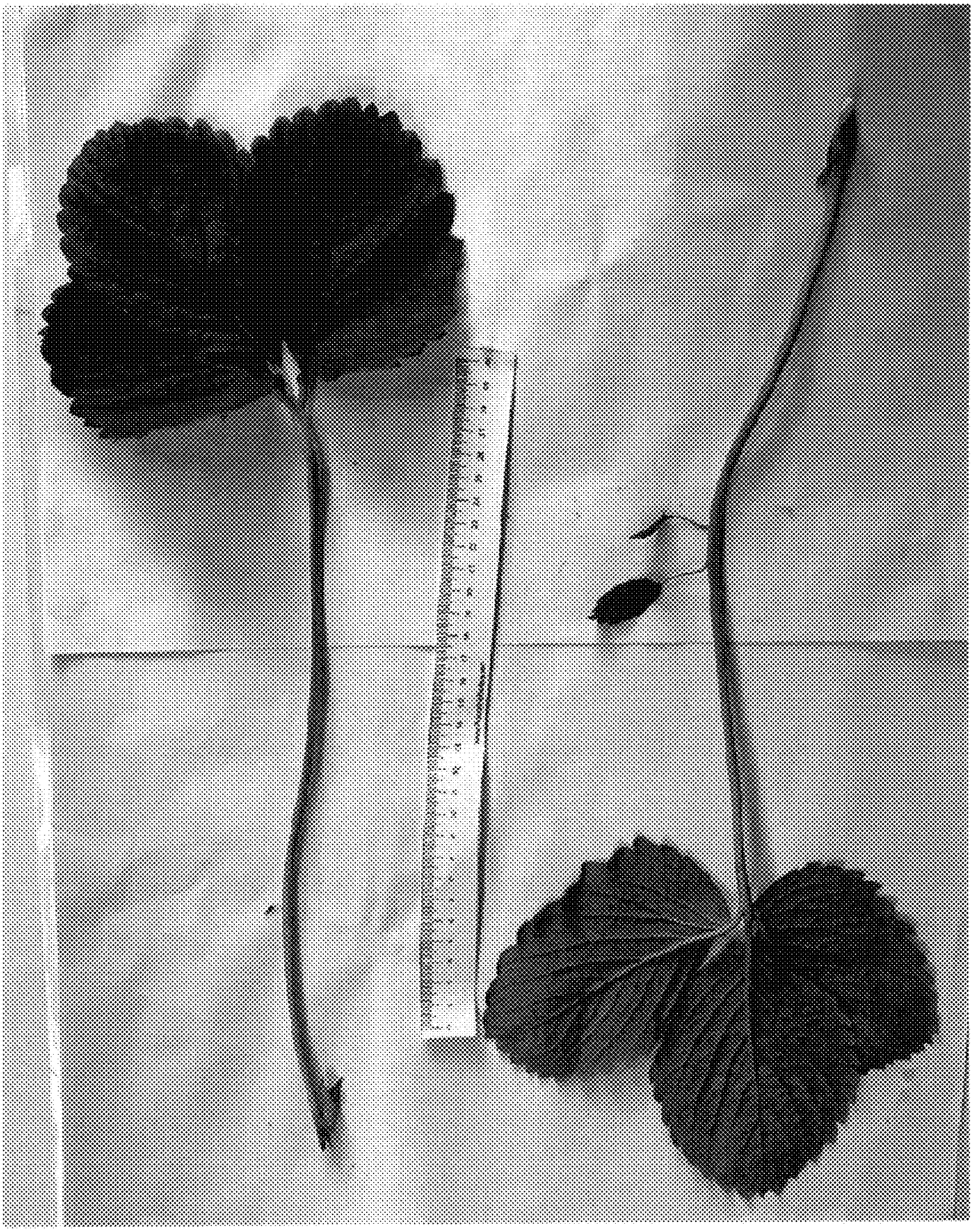

FIG. 4

FIG. 5